

An abstract graphic on the left side of the slide. It features a dark background with glowing, stylized circuitry and data lines. The lines are primarily green and blue, with some red and orange accents. They flow and curve through the space, resembling a complex network or a futuristic circuit board. The overall aesthetic is high-tech and digital.

Module 01 - LLM Behavior Analysis & Evaluation Techniques

Building the Foundation for AI Engineering Excellence

Understanding Transformers: The Foundation of Modern LLMs

A high-level roadmap to the core concepts powering large language models — preparing you for a deeper technical dive into mechanistic interpretability and attention mechanisms.



Why This Session Matters

The Big Picture First

Before we dissect the intricate mechanics of transformers, we need to understand the landscape. This session establishes the conceptual framework that makes advanced topics like attention heads, ablations, and interpretability meaningful.

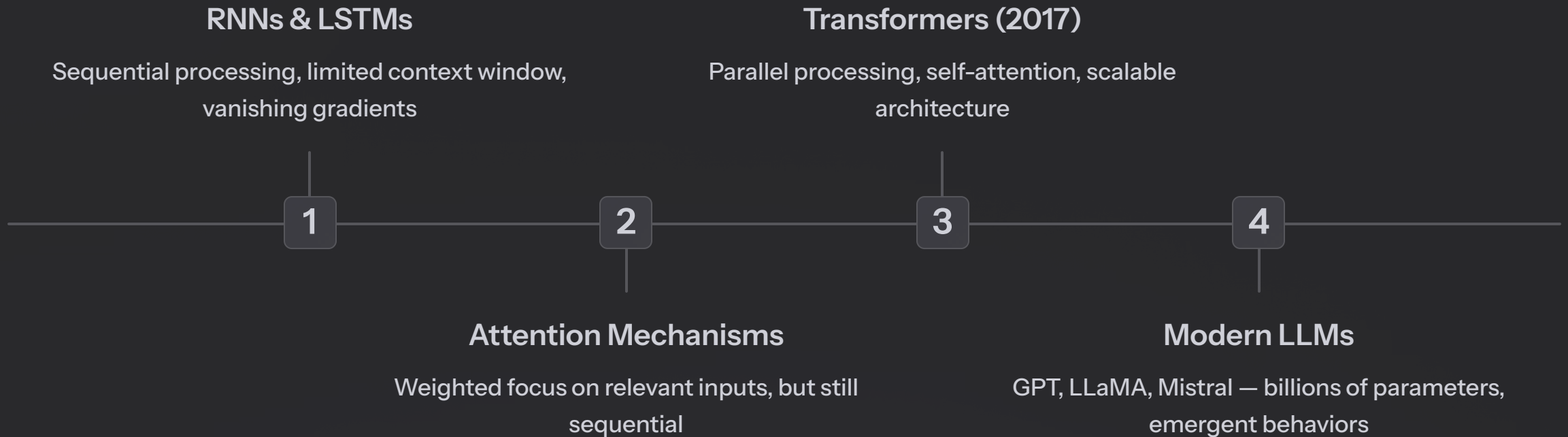
Think of this as your architectural blueprint before we examine the individual components under a microscope.

Setting the Stage

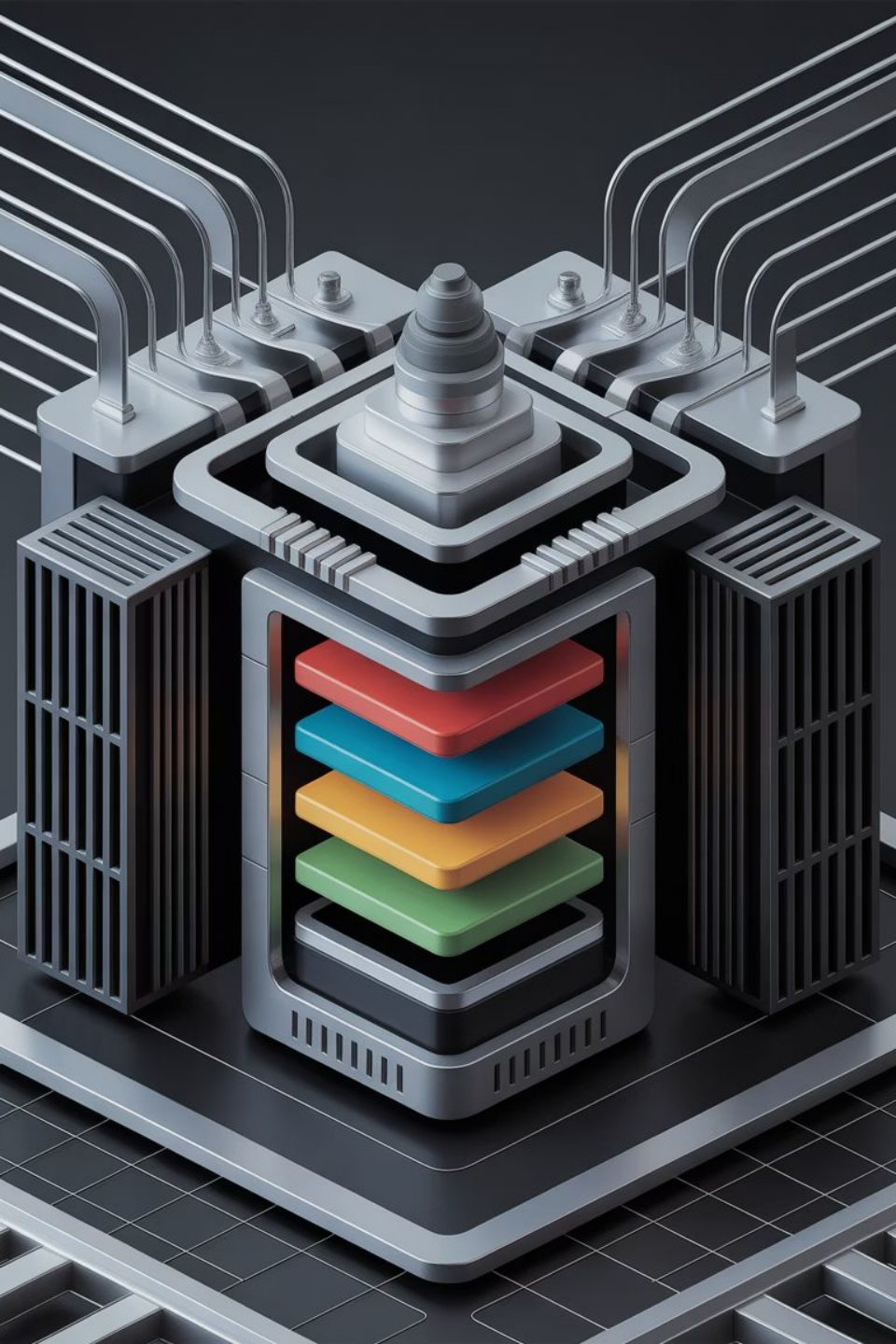
You'll learn where transformers fit in the AI evolution, why attention mechanisms matter, and how components like queries, keys, and values work together at a high level.

This foundation will make your deep-dive session far more intuitive and actionable.

The Evolution to Transformers



The transformer architecture revolutionized NLP by solving the sequential bottleneck. Unlike RNNs that process tokens one-by-one, transformers analyze entire sequences simultaneously through attention mechanisms. This parallelization enabled the scale we see in today's LLMs.



The Transformer Architecture at a Glance

Transformers consist of stacked layers, each containing attention mechanisms and feed-forward networks. Input tokens are converted to embeddings, processed through multiple layers that refine representations, and finally decoded into predictions.

The key innovation: **self-attention** lets each token draw information from other tokens in the sequence, capturing long-range dependencies that earlier architectures struggled with. In the decoder-only models used for text generation (GPT, LLaMA), a token attends only to itself and the tokens before it, a causal mask that is exactly what makes left-to-right generation work. Bidirectional encoders like BERT instead attend across the full sequence.

What Makes Attention "Attention"?

The Core Problem

How does a model decide which parts of the input are relevant for predicting the next token?

In the sentence "The cat sat on the mat because it was comfortable," what does "it" refer to?

The Attention Solution

Attention mechanisms compute relevance scores between every pair of tokens, allowing the model to dynamically focus on important context.

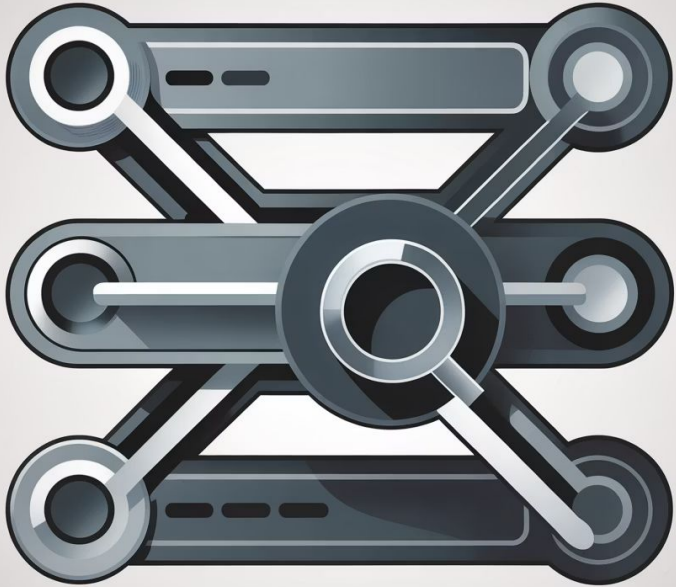
For "it," the model learns to attend strongly to "mat" based on semantic relationships.

Why It's Powerful

Unlike fixed-window approaches, attention adapts to each specific input, learning which relationships matter through training.

This flexibility enables sophisticated language understanding and generation.

The Query-Key-Value Framework: A High-Level Intuition



Think of it Like a Search Engine

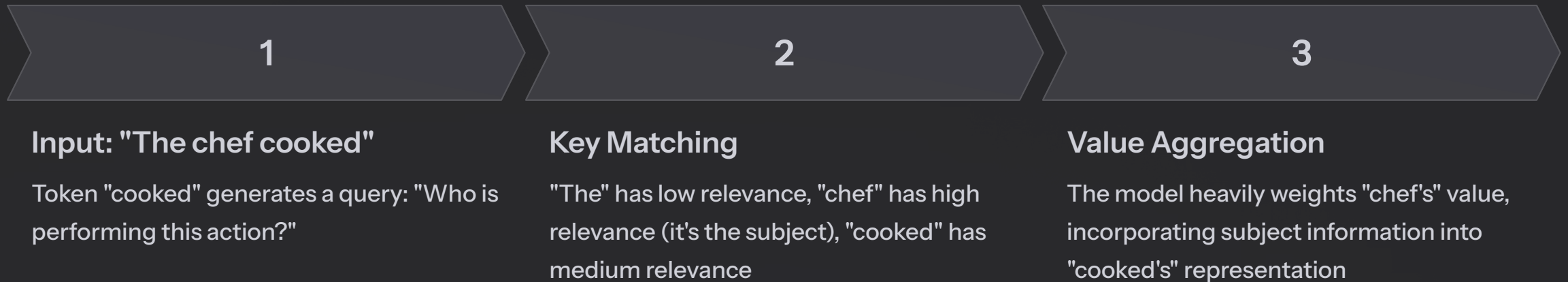
Query: "What am I looking for?" — Each token generates a query representing what information it needs.

Key: "What do I contain?" — Each token advertises what information it can provide.

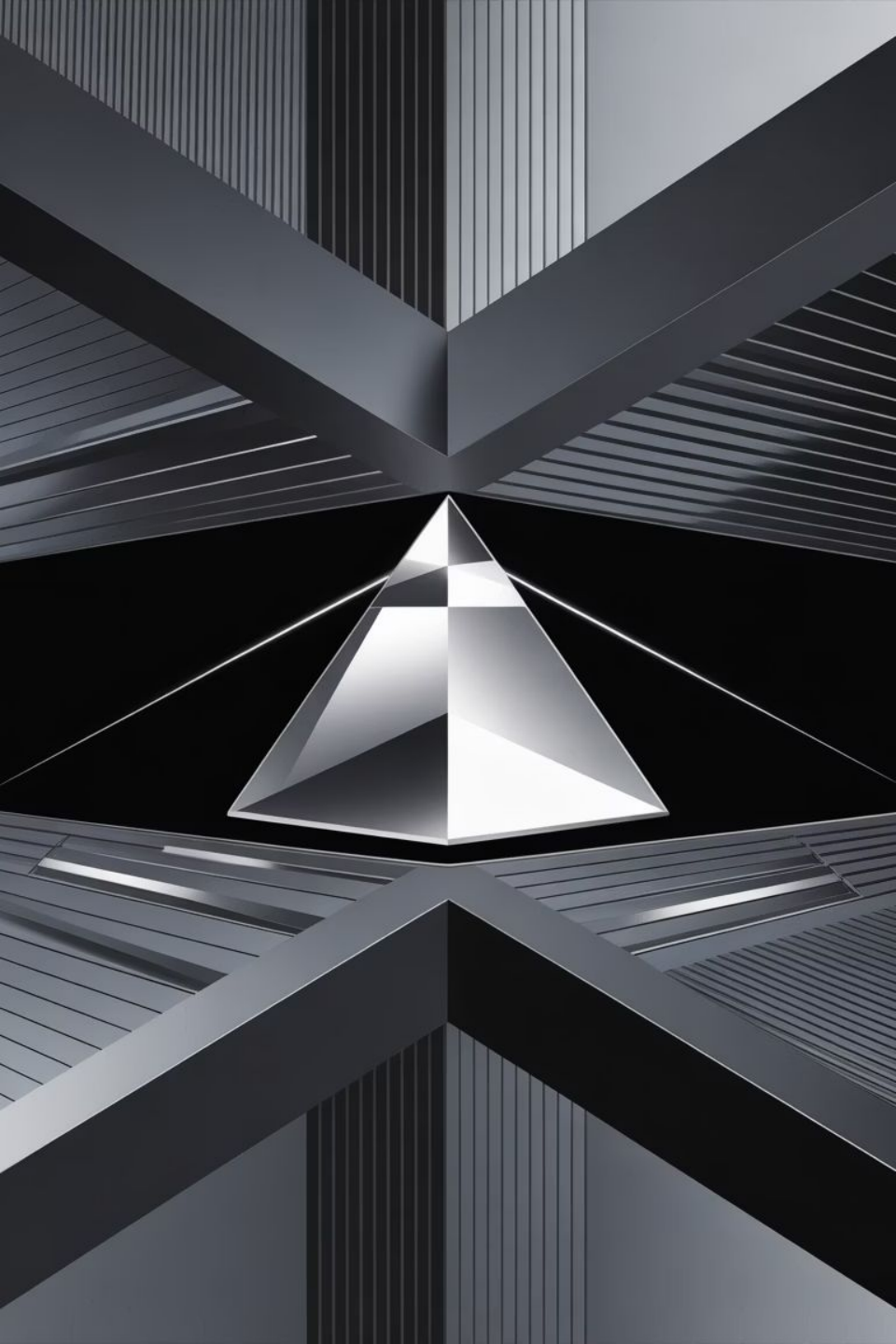
Value: "Here's my actual content" — The information that gets passed along when there's a match.

The model compares queries against all keys, computes attention scores (how relevant each key is to each query), then aggregates values weighted by those scores.

QKV in Action: A Concrete Example



This process happens in parallel across all token positions and is repeated in multiple attention heads, each potentially learning different linguistic patterns.



Multi-Head Attention: Multiple Perspectives

A single attention mechanism might miss important patterns. Multi-head attention runs several attention operations in parallel, each with different learned parameters. Think of it as multiple experts examining the same text from different angles.

One head might specialize in subject-verb relationships, another in pronoun resolution, another in semantic similarity. The outputs are concatenated and transformed, giving the model a richer, multi-faceted understanding of the input.

Why Multiple Heads Matter



Syntactic Patterns

Some heads learn grammatical structures: subject-verb agreement, dependency parsing, phrase boundaries.



Positional Information

Certain heads track relative positions, helping the model understand word order and sequence structure.



Semantic Relationships

Other heads focus on meaning: synonyms, analogies, conceptual similarity, topic coherence.



Copying Mechanisms

Some heads learn to simply copy or move information from one position to another, essential for tasks like name recall.

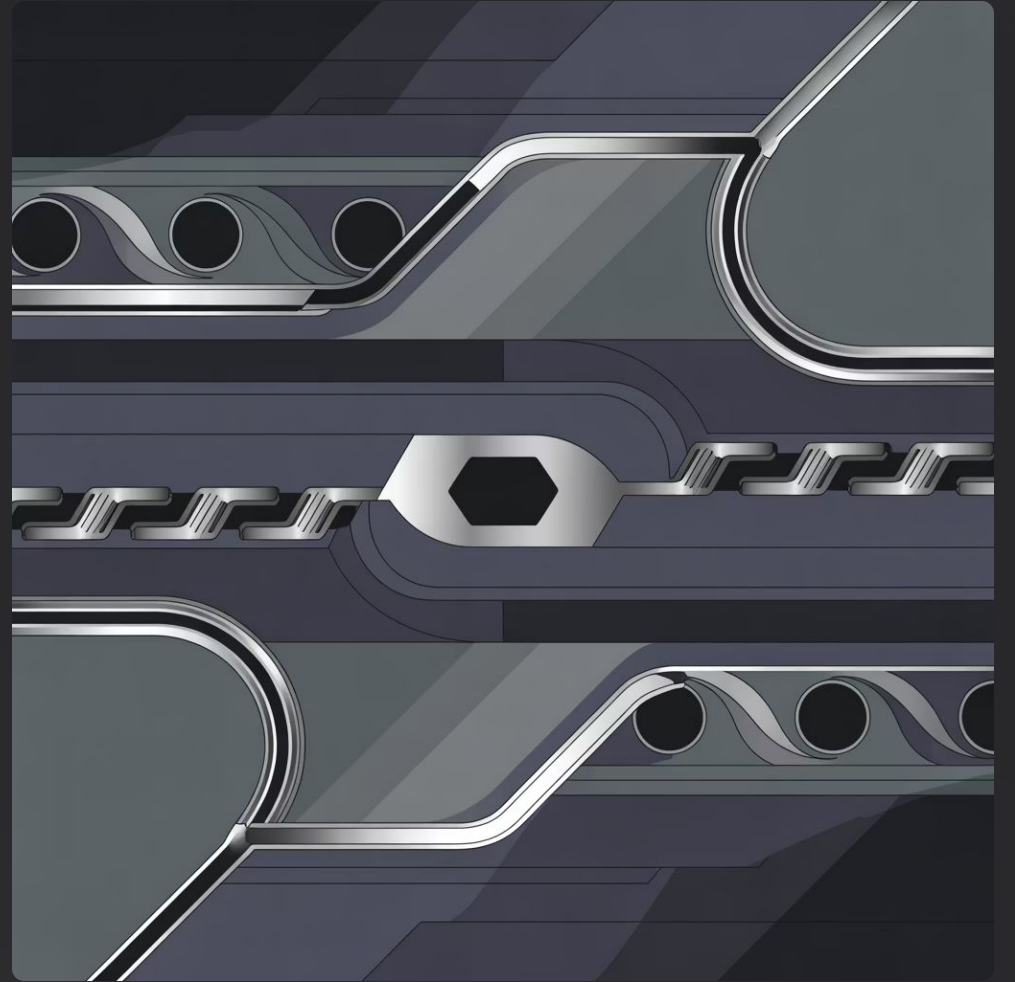
Specialized Attention Heads: Induction Heads

Pattern Completion Specialists

Induction heads are attention heads that learn to recognize and complete repeated patterns. They're crucial for in-context learning — the ability to pick up on patterns within a single prompt.

Example: Given a repeated sequence like "... K7 Qz ... K7", an induction head predicts "Qz" by matching the current token to its earlier occurrence and copying what followed it last time. This prefix-match-and-copy is the core of in-context learning.

Induction heads are implemented by a two-layer circuit: a previous-token head in an earlier layer tags each position with its predecessor, and an induction head in a later layer uses those tags to find the match. They form at a distinct phase change during training and are foundational for more complex reasoning, demonstrating that transformers learn generalizable algorithms, not just memorized text.



Specialized Attention Heads: Name Mover Heads

01

Information Retrieval

Name mover heads specialize in retrieving specific information, particularly proper nouns and factual details, from earlier in the context.

02

How They Work

When the model needs to recall a name or fact mentioned earlier, these heads attend back to the relevant token and "move" that information forward to the current position.

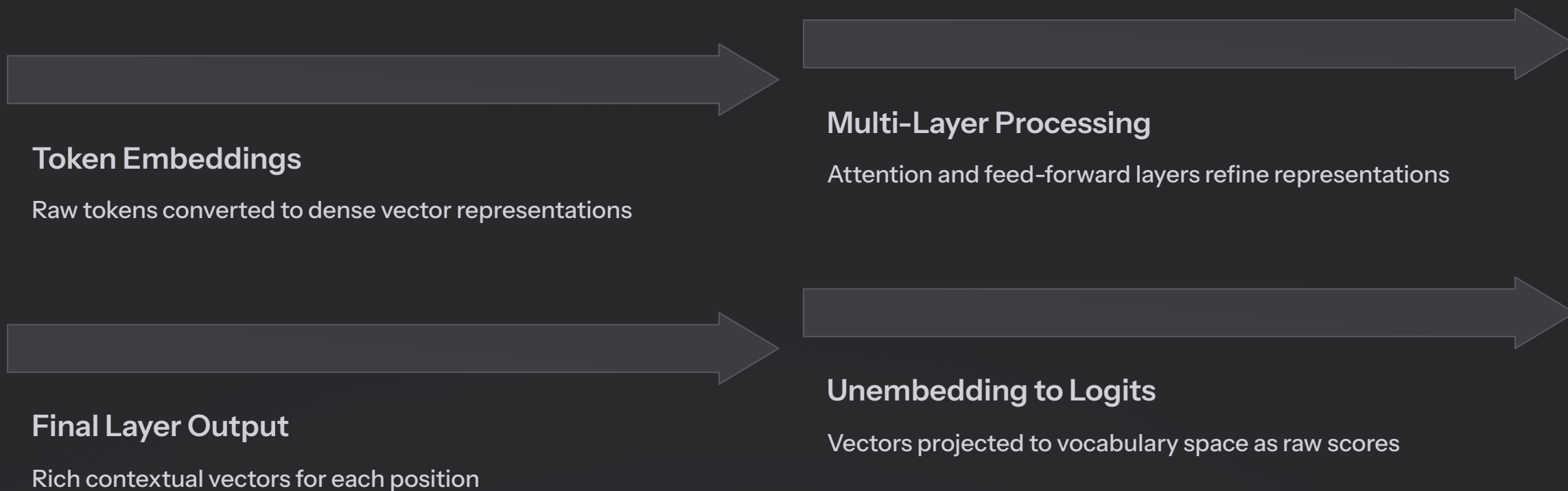
03

Why They're Important

Name mover heads are essential for coherent long-form generation, question answering, and maintaining consistency across extended contexts.

Understanding these specialized heads is key to mechanistic interpretability — knowing not just that a model works, but *how* and *why* specific components contribute to behavior.

The Path from Attention to Predictions





Understanding Logits: The Model's "Thoughts"

Logits are the raw, unnormalized scores the model assigns to each token in its vocabulary before making a prediction. For a vocabulary of 100,000-plus tokens (Llama 3 uses about 128,000; the GPT-4o and o-series tokenizers use about 200,000), you get that many logit values at every prediction step. Note these are subword tokens, not whole words.

Higher logits indicate stronger "belief" that a particular token should come next. These scores are then converted to probabilities via softmax, but examining raw logits gives us insight into the model's internal preferences before that normalization.

From Logits to Predictions



Raw Logits

Unbounded scores: "cat" = 3.0, "dog" = 2.75, "bird" = 2.08, "the" = -0.11



Softmax Normalization

Convert to probabilities: "cat" = 45%, "dog" = 35%, "bird" = 18%, "the" = 2%



Sampling Strategy

Top-k, nucleus sampling, or temperature to introduce controlled randomness



Token Selection

Final prediction based on probability distribution and sampling parameters

Why Logits Matter for Interpretability

Direct Model Insight

Logits reveal what the model "thinks" before sampling randomness is applied. By analyzing logit differences, we can understand which components (heads, layers, neurons) contribute most to specific predictions.

For instance, if removing an attention head drops the logit for "Paris" when answering "Capital of France?", that head is clearly important for that factual recall.

Causal Analysis Foundation

Logit analysis is the basis for techniques like **logit attribution** and **activation patching**. We can trace how information flows through the network by observing logit changes.

This forms the bridge between architecture and behavior — connecting the mechanics to the output.



Ablation Studies: The Scientific Method for Transformers

Ablation means systematically removing or modifying components to understand their causal role. In neuroscience, researchers ablate brain regions; in mechanistic interpretability, we ablate attention heads, layers, or individual neurons.

By observing how model performance degrades (or doesn't) when we "knock out" a component, we learn what that component actually does. This moves us beyond correlation to causation.

Types of Ablations in Practice

1

Zero Ablation

Set attention head outputs or neuron activations to zero. Simple but can create distribution shifts the model never saw during training.

2

Mean Ablation

Replace activations with their average value across a dataset. More realistic, reduces artifacts from out-of-distribution behavior.

3

Resample Ablation

Replace with activations from a different input. Tests whether the component is encoding input-specific vs. general information.

4

Activation Patching

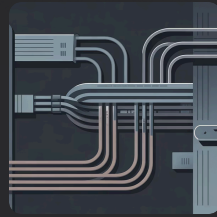
Replace activations from a "corrupted" input with those from a "clean" input to localize where information is processed.

What Ablations Reveal



Component Importance

If ablating a head barely affects performance, it's not critical for that task. Large performance drops indicate important components.



Redundancy Patterns

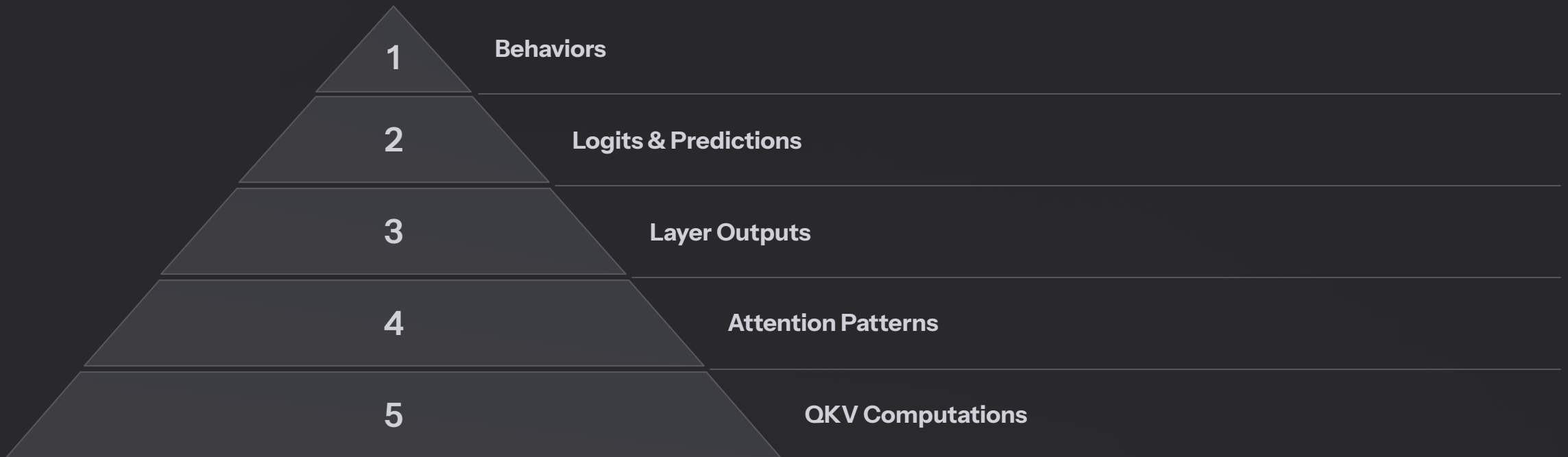
Multiple heads often learn similar functions. Ablating one may have little effect due to redundancy, but ablating several simultaneously reveals their collective role.



Information Flow

By systematically ablating across layers, we trace how information is transformed and refined as it moves through the network.

Connecting the Concepts: The Interpretability Stack

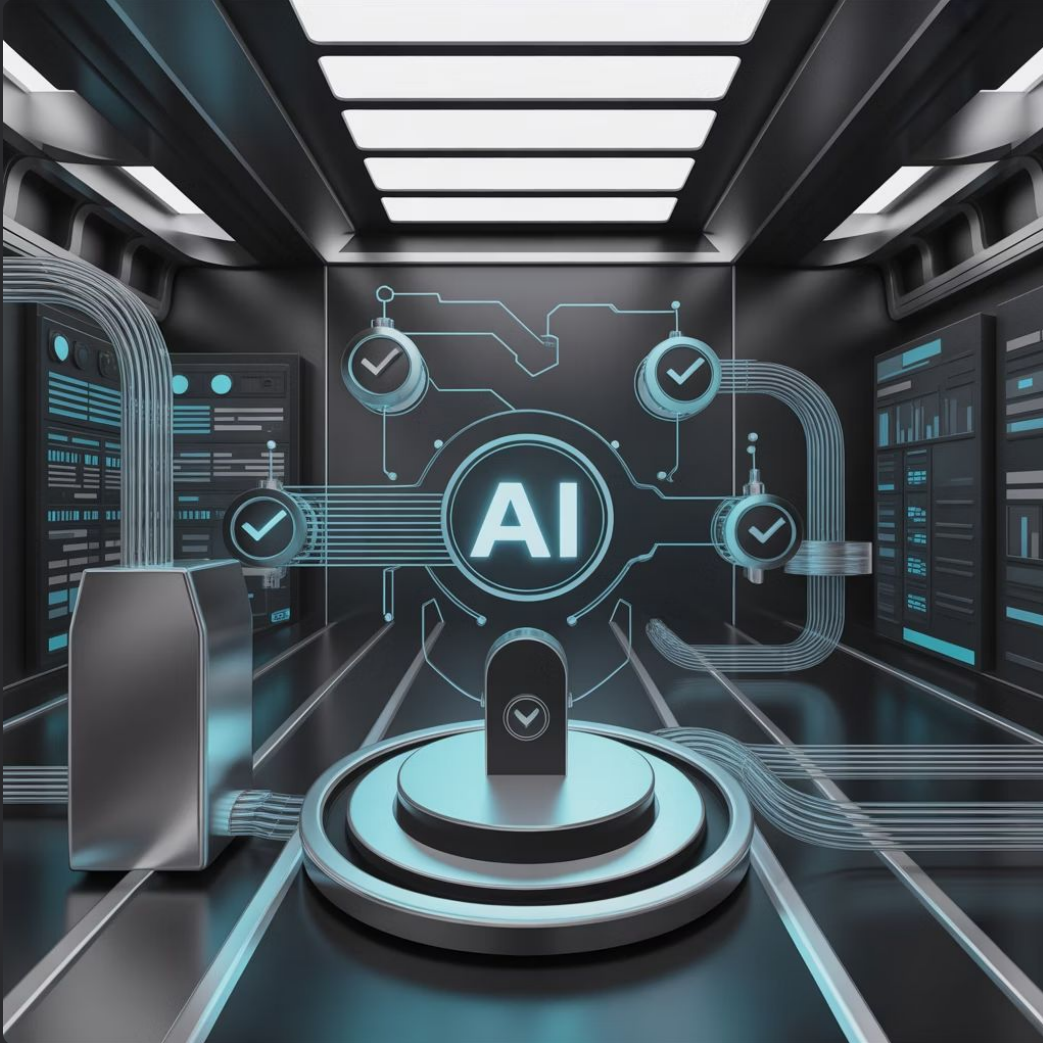


Each layer of abstraction builds on the ones below. Mechanistic interpretability works by connecting high-level behaviors down to low-level computations, and ablations help us establish those causal links.

Where This Fits in Your Learning Journey



Real-World Implications: Why This Matters



Beyond Academic Curiosity

AI Safety: Understanding how models work helps identify failure modes, biases, and vulnerabilities before deployment.

Debugging & Improvement: When a model behaves unexpectedly, interpretability tools help diagnose the root cause rather than guessing.

Alignment & Control: To build AI systems that reliably do what we want, we need to understand their internal reasoning processes.

Mechanistic interpretability is foundational to responsible AI development at scale.

Key Terminology Quick Reference

Transformer

Neural network architecture using self-attention for parallel sequence processing

Attention

Mechanism for weighting relevance of different input tokens dynamically

Query/Key/Value

Three projections enabling attention: what to look for, what's available, what to retrieve

Multi-Head

Running multiple attention mechanisms in parallel with different parameters

Induction Head

Specialized attention head learning to complete repeated patterns

Name Mover Head

Attention head retrieving specific factual information from context

Logits

Raw unnormalized scores for each vocabulary token before prediction

Ablation

Removing or modifying components to test their causal importance

Preparing for the Deep Dive

1

You Now Understand the Landscape

You know where transformers came from, why attention matters, and how components fit together conceptually.

2

You Have the Vocabulary

Terms like QKV, logits, ablation, and attention heads are no longer abstract — you understand their purpose and relationships.

3

You're Ready for Tools

The next session introduces TransformerLens and hands-on techniques. With this foundation, those tools will make immediate sense.



Questions to Carry Forward

As you move into the deep dive, keep these questions in mind:

- Which specific attention heads are most important for the task I care about?
- How do different heads interact and compose to produce complex behaviors?
- What ablation strategy best isolates the causal effect I want to measure?
- How can I visualize attention patterns to build intuition about what's happening?
- What are the failure modes and limitations of interpretability techniques?

These questions will guide your hands-on exploration in the next session.

Ready for the Deep Dive

You've Built the Foundation

You understand the transformer architecture, attention mechanisms, specialized heads, logits, and ablation techniques at a conceptual level. This roadmap positions you to dive deep into TransformerLens, attention saliency, and mechanistic interpretability with confidence.

The tools and techniques you'll encounter next will connect directly to the concepts we've covered here. Let's move from theory to practice.

